

The Flaws of Others

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Abstract

1 Introduction

Large Language Models (LLMs) are advanced artificial intelligence systems trained on vast amounts of text data to generate human-like language. These models utilize deep learning techniques to understand and produce text based on the patterns and structures found in their training data [Vaswani et al., 2017, Brown et al., 2020]. LLMs have demonstrated impressive capabilities in various natural language processing tasks, including text generation, translation, and question-answering.

Despite their remarkable performance, LLMs are prone to generating false or misleading statements, a phenomenon often referred to as "hallucinations" [Ji et al., 2022]. Hallucinations in LLMs can manifest as factual inconsistencies, logical contradictions, or entirely fabricated content that appears plausible [Maynez et al., 2020]. This issue is exacerbated by the lack of a verification mechanism within the models themselves [Bender et al., 2021, Weidinger et al., 2021].

Several factors contribute to the occurrence of hallucinations, including biases in training data, limitations in knowledge representation, and the models' tendency to prioritize fluency over factual accuracy [Lin et al., 2022]. The prevalence and impact of hallucinations are significant, with quantitative evaluations revealing that they occur in up to 30% of generated responses, substantially affecting the trustworthiness of these models [Ji et al., 2022, Lin et al., 2022]. Moreover, studies have shown that LLMs can generate false information even when explicitly prompted to be truthful [Evans et al., 2021], underscoring the challenge of aligning model outputs with factual correctness.

To address this problem, researchers have developed various mitigation strategies. These include techniques such as retrieval-augmented generation [Lewis et al., 2020], which aims to ground model responses in external knowledge sources, and self-consistency checks [Wang et al., 2023], which attempt to improve the internal coherence of model outputs. Despite these efforts, hallucinations remain a significant challenge in the development of reliable AI systems.

The persistence of this issue highlights the need for ongoing research and development of robust verification methods. As LLMs continue to evolve and find applications in diverse

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fields, ensuring their factual accuracy and reliability becomes increasingly crucial. This challenge sits at the intersection of natural language processing, knowledge representation, and AI ethics, requiring interdisciplinary approaches to develop comprehensive solutions.

1.1 Invalidation vs. Hallucination

While hallucinations in LLMs have been extensively studied, this paper introduces a novel concept called “invalidation,” which represents a distinct and potentially more insidious form of false information generation. Invalidation differs from hallucination in several key aspects:

1. Nature of Falsehood: Hallucinations are unintentional factual inaccuracies or fabrications, whereas invalidation involves the active negation or disputation of established truths.
2. Presentation of Information: Hallucinations typically result in plausible but incorrect information. In contrast, invalidation often presents alternative facts or misleading evidence that directly contradicts known truths.
3. Impact on Knowledge Systems: Hallucinations may lead to isolated errors, but invalidation has the potential to undermine entire systems of knowledge by creating doubt and confusion about verified facts.
4. Underlying Mechanisms: Hallucinations stem from the model’s probabilistic nature and limitations in training data. Invalidation, while not intentional, appears more deliberate due to the emergent behavior of the model and the influence of biased or misleading training data.

For example, a hallucination might involve an LLM incorrectly stating the year of a historical event. Invalidation, on the other hand, might manifest as the LLM generating content that denies well-established scientific facts, such as climate change, by presenting fabricated data or misinterpreting genuine information.

This distinction between hallucination and invalidation is crucial for several reasons:

1. It allows for more targeted mitigation strategies, as the approaches needed to address invalidation may differ from those used for hallucinations.
2. It highlights a potentially more dangerous form of misinformation that could have far-reaching consequences in fields such as science, politics, and public health.
3. It emphasizes the need for more sophisticated verification mechanisms that can not only check factual accuracy but also detect attempts to undermine established knowledge.

To comprehensively understand and address these phenomena, we propose an epistemological framework that categorizes false information generated by LLMs into two distinct types: hallucination and invalidation. This framework helps delineate the nature, causes, and implications of each type, thus guiding the development of targeted mitigation strategies.

1.2 Epistemological Framework for False Information

1. Hallucination: Characterized by unintentional factual inaccuracies and fabrications due to the probabilistic text generation process. Mitigation strategies include improving training data quality, enhancing model architectures to prioritize factual accuracy, and incorporating real-time fact-checking mechanisms [Lewis et al., 2020, Wang et al., 2023]. For example, an LLM might generate a fabricated historical event due to the patterns it learned from the training data.

2. **Invalidation:** Defined by statements that negate or contradict established facts through misleading narratives or false evidence. Addressing invalidation requires methods to cross-verify generated content with credible sources and detect contradictions with known truths. Techniques like retrieval-augmented generation and consistency checks can be effective [Evans et al., 2021]. For instance, generating false data to deny climate change exemplifies invalidation.

By distinguishing between these two types of false information, this framework not only aids in understanding the underlying mechanisms but also informs the development of more precise and effective countermeasures. Such an approach is critical in enhancing the reliability and trustworthiness of AI-driven language models, particularly as they become increasingly integrated into various domains of society. This framework also highlights important considerations in the broader context of AI ethics and the ongoing development of LLMs.

To address the problem of invalidation, some approaches suggest using multiple LLMs to cross-verify information [Li et al., 2021]. By generating a consensus among different models, it is possible to reduce the likelihood of false statements. This method leverages the diverse training backgrounds and potential error distributions of multiple models, enhancing the overall accuracy and reliability of the generated content.

1.3 Invalidation in Human Discourse and Its Universal Implications

Invalidation, as observed in LLMs, is not an isolated phenomenon unique to artificial intelligence. Rather, it mirrors a well-documented pattern in human discourse, suggesting a potentially universal cognitive process that transcends the boundaries between artificial and human intelligence.

In social interactions, humans frequently encounter and sometimes generate statements that contradict observable evidence or present fabricated narratives as facts. This phenomenon manifests in various forms, from the denial of scientifically proven phenomena to the propagation of conspiracy theories. The parallel between human and LLM behavior in this regard is striking and warrants deeper exploration.

Sociological perspectives offer valuable insights into this phenomenon. Goffman [1959] illuminates how individuals shape and present narratives that may invalidate certain states of the world, a process strikingly similar to the way LLMs construct alternative realities. Festinger [1957] theory of cognitive dissonance explains how people manage conflicting information and beliefs, often leading to the dismissal of evidence that contradicts their views - a process that may be analogous to how LLMs prioritize internal consistency over factual accuracy. Cohen [2001] work on states of denial further illustrates how groups may collectively reject realities such as human rights violations, mirroring the way LLMs can generate content that negates established facts.

The field of media studies and communication theory provides additional parallels. McLuhan [1964] insights into how different media forms influence the dissemination and acceptance of information, including falsehoods, can be extended to understand how the architecture of LLMs shapes their output. Chomsky and Herman [1988] analysis of how mass media shapes public perception by filtering and framing information resonates with the way LLMs process

and generate content based on their training data. Gerbner et al. [1976] cultivation theory, examining how prolonged exposure to media content can alter perceptions of reality, offers a framework for understanding how repeated exposure to certain patterns in training data might lead LLMs to generate invalidating content.

Political science and propaganda studies further illuminate this phenomenon. Arendt [1951] discussions on the use of propaganda to manipulate public perception find echoes in the way LLMs can generate misleading narratives. Ellul [1965] detailed analysis of how propaganda shapes beliefs and attitudes, leading to the invalidation of certain truths, provides a lens through which we can understand the potential societal impact of LLM-generated invalidations.

From a cognitive science perspective, Kahneman and Tversky [1974] work on heuristics and biases offers a framework for understanding why both humans and LLMs may accept and generate false narratives. Pinker [2002] exploration of how ideological beliefs can lead to the rejection of scientific evidence provides insight into the mechanisms behind invalidation in both human and artificial systems.

The striking parallels between invalidation in human discourse and in LLMs point to a potentially universal phenomenon rooted in the fundamental processes of information processing and knowledge representation. This similarity suggests that invalidation may not be merely a flaw in AI systems, but rather a manifestation of broader cognitive patterns that emerge in any sufficiently complex information processing system, whether biological or artificial.

Understanding invalidation as a universal phenomenon has profound implications for AI development and our broader understanding of cognition. It suggests that addressing invalidation in LLMs may require insights from diverse fields such as sociology, psychology, and cognitive science, in addition to computer science. Moreover, studying how LLMs produce invalidations may provide new perspectives on human cognitive processes, potentially offering insights into phenomena like confirmation bias, belief perseverance, and the spread of misinformation in human societies.

1.4 Goal

In this paper, we... [state the focus and contribution of the manuscript... to be done last based on all pieces that can be harmonized]

2 Methods

2.1 Formalization of Invalidation

To systematically study the phenomenon of invalidation, we propose a formal model that quantifies how invalidations propagate within a discursive network. This model considers actors as nodes in a network, with edges representing the exchange of statements. The goal is to understand how actors influence each other and how invalidations spread, eventually reaching an equilibrium state. The subsequent section will detail the methods used to develop

and analyze this model, providing a quantitative framework for studying invalidation in both human and artificial contexts.

2.2 Basic Invalidation Propagation

Definition 2.1. Actors and Statements. Let $A = \{a_1, a_2, \dots, a_n\}$ be the set of actors in the discursive network, and let $S = \{s_1, s_2, \dots, s_m\}$ be the set of all possible statements in the network, where each statement s_i can be either true or false.

Proposition 2.1. *Each actor a_i holds a set of beliefs, represented by $B_i \subseteq S$. A belief is a statement that the actor considers true. The goal function G_i , $G_i : A \rightarrow 2^S$, the power set of S , represents the set of statements a_i aims to convince a_j to believe. This includes every combination of elements that can be formed from S , ranging from the empty set to the full set S itself.*

Proposition 2.2. *Communication between actors involves the exchange of statements. A communication from actor a_i to actor a_j is represented as $C_{ij} = \{s_k \mid s_k \in B_i\}$, where s_k is a statement that a_i believes and wants a_j to accept. Invalidation occurs when an actor presents a statement that directly contradicts another statement or fabricates evidence to support a false proposition. Formally, $I_{ij}(s_k, s_l)$ denotes that actor a_i invalidates statement s_k held by a_j by presenting s_l .*

Proposition 2.3. *The persuasion function $P_{ij}(s_k)$ represents the likelihood that actor a_j will adopt statement s_k after communication from a_i . This can be modeled using probabilities or influence weights. The update rule $U_j(B_j, C_{ij}, I_{ij})$ determines how the belief set B_j of actor a_j is updated after receiving communication C_{ij} from actor a_i and experiencing invalidation I_{ij} .*

Proposition 2.4. *Let $N = (A, S, B, G, C, I, P, U)$ be the formal model of the discursive network, where A is the set of actors, S is the set of statements, $B = \{B_1, B_2, \dots, B_n\}$ is the set of belief sets for each actor, $G = \{G_1, G_2, \dots, G_n\}$ is the set of goal functions for each actor, $C = \{C_{ij} \mid i, j \in \{1, 2, \dots, n\}\}$ is the set of communications between actors, $I = \{I_{ij} \mid i, j \in \{1, 2, \dots, n\}\}$ is the set of invalidations between actors, $P = \{P_{ij} \mid i, j \in \{1, 2, \dots, n\}\}$ is the set of persuasion functions, and $U = \{U_j \mid j \in \{1, 2, \dots, n\}\}$ is the set of update rules.*

Example. Consider a simple scenario with three actors $A = \{a_1, a_2, a_3\}$ and two statements $S = \{s_1, s_2\}$. Actor a_1 believes s_1 ($B_1 = \{s_1\}$) and wants a_2 and a_3 to also believe s_1 ($G_1(a_2) = G_1(a_3) = \{s_1\}$). Actor a_2 believes s_2 ($B_2 = \{s_2\}$) and wants a_1 and a_3 to believe s_2 ($G_2(a_1) = G_2(a_3) = \{s_2\}$). Actor a_3 is initially neutral ($B_3 = \emptyset$). Actor a_1 communicates $C_{12} = \{s_1\}$ to a_2 , who invalidates s_1 by presenting $I_{21}(s_1, s_2)$. Actor a_3 , observing this interaction, updates their belief set based on the persuasion functions and update rules. This framework models the propagation of invalidation within a discursive network, capturing the dynamics of belief, communication, and influence. By formalizing these interactions, we can analyze and predict how invalidation affects the acceptance and rejection of statements among actors in the network.

Proposition 2.5. Basic Model Formulation. *To describe the state of asymptotic equilibrium of the system mathematically, we need to analyze the transition probabilities between the states s_1 and s_2 . This can be modeled using a Markov chain with two states.*

Define the states as follows: S_1 : The state where an actor believes s_1 . S_2 : The state where an actor believes s_2 . The transition probabilities are: p_{12} : Probability that an actor transitions from S_1 to S_2 , and p_{21} : Probability that an actor transitions from S_2 to S_1 . Let $P(t)$ represent the state vector at time t , where: $P_1(t)$: Proportion of actors in state S_1 at time t and $P_2(t)$: Proportion of actors in state S_2 at time t .

The transition matrix T is given by:

$$T = \begin{pmatrix} 1 - p_{12} & p_{12} \\ p_{21} & 1 - p_{21} \end{pmatrix} \quad (1)$$

The state vector $P(t)$ evolves according to:

$$P(t + 1) = T \cdot P(t) \quad (2)$$

Theorem 2.1. Basic Invalidation Propagation. *The state of equilibrium of the system described by Equation 2 is a stable fixed point given by:*

$$P = \begin{pmatrix} \frac{p_{21}}{p_{12} + p_{21}} \\ \frac{p_{12}}{p_{12} + p_{21}} \end{pmatrix}$$

Proof. The system reaches equilibrium when $P(t + 1) = P(t) = P$, where P is the equilibrium state vector. At equilibrium, the proportions of actors holding each belief remain constant. We need to solve the equation:

$$P = T \cdot P. \quad (3)$$

This can be written as a system of linear equations:

$$\begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = \begin{pmatrix} 1 - p_{12} & p_{21} \\ p_{12} & 1 - p_{21} \end{pmatrix} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}, \quad (4)$$

or in algebraic form

$$P_1 = (1 - p_{12})P_1 + p_{21}P_2, \quad (5)$$

$$P_2 = p_{12}P_1 + (1 - p_{21})P_2. \quad (6)$$

Rearranging, we obtain

$$P_1 = P_1 - p_{12}P_1 + p_{21}P_2, \quad (7)$$

$$P_2 = p_{12}P_1 + P_2 - p_{21}P_2. \quad (8)$$

Simplifying,

$$-P_1p_{12} + P_2p_{12} = 0 \quad (9)$$

$$P_1p_{12} = P_2p_{12} \quad (10)$$

Since $p_{12} \neq 0$:

$$P_1 = P_2 \cdot \frac{p_{21}}{p_{12}} \quad (11)$$

From the normalization condition $P_1 + P_2 = 1$:

$$P_2 \cdot \frac{p_{21}}{p_{12}} + P_2 = 1 \quad (12)$$

from where

$$P_2 = \frac{p_{12}}{p_{12} + p_{21}} \quad (13)$$

Substitute P_2 back into the equation for P_1 :

$$P_1 = \frac{p_{21}}{p_{12}} \cdot P_2 \quad (14)$$

$$P_1 = \frac{p_{21}}{p_{12} + p_{21}} \quad (15)$$

Thus, the equilibrium state vector is:

$$P = \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = \begin{pmatrix} \frac{p_{21}}{p_{12} + p_{21}} \\ \frac{p_{12}}{p_{12} + p_{21}} \end{pmatrix}. \quad (16)$$

Since system 2 is linear, its stability can be assessed by determining the values of the roots of the characteristic polynomial resulting from assessing $\det(T - \lambda I) = 0$. The characteristic polynomial is

$$\lambda^2 + (p_{12} + p_{21} - 2) \lambda + 1 - p_{12} - p_{21} = 0,$$

resulting in the roots $\{1, 1 - p_{12} - p_{21}\}$. The stability of the first root is undecidable, but the second root is clearly less than one, which is the necessary and sufficient condition for stability for discrete dynamical systems. Thus, the fixed point is stable. This completes the proof. $\square$

Interpretation: The proportion of actors holding belief s_1 is P_1 and the proportion of actors holding belief s_2 is P_2 . When $p_{12} > p_{21}$, i.e., when the probability of transitioning from s_1 to s_2 is higher than the probability of transitioning from s_2 to s_1 , more actors will shift their belief from s_1 to s_2 than from s_2 to s_1 over time.

P_1 (**Proportion of actors holding belief s_1**): $P_1 = \frac{p_{21}}{p_{12} + p_{21}}$. Since $p_{12} > p_{21}$, P_1 will be less than 0.5. This means that fewer actors will hold belief s_1 at equilibrium compared to belief s_2 .

P_2 (**Proportion of actors holding belief s_2**): $P_2 = \frac{p_{12}}{p_{12} + p_{21}}$. Since $p_{12} > p_{21}$, P_2 will be greater than 0.5. This means that more actors will hold belief s_2 at equilibrium.

In summary, when $p_{12} > p_{21}$, the system will reach an equilibrium where the proportion of actors holding the belief s_2 is higher than the proportion of actors holding the belief s_1 . This reflects the higher likelihood of actors transitioning from s_1 to s_2 compared to the reverse.

2.3 Cross-Network Invalidation Detection Model

We present an enhanced model of two interconnected discursive networks, each with the capacity to generate falsehoods internally and detect falsehoods in the other network. This refined model incorporates time-dependent parameters, accounts for network size effects, and allows for asymmetry between networks.

Definition 2.2. Dual Networks. Let N_1 and N_2 be two discursive networks, with sets of actors $A_1 = \{a_{11}, a_{12}, \dots, a_{1n_1}\}$ and $A_2 = \{a_{21}, a_{22}, \dots, a_{2n_2}\}$ respectively, where n_1 and n_2 are the number of actors in each network.

Proposition 2.6. Network State. *The state of each network N_k at time t is represented by $S_k(t) = (T_k(t), F_k(t))$, where $T_k(t)$ is the number of true statements and $F_k(t)$ is the number of false statements in network N_k at time t .*

Proposition 2.7. Falsehood Generation. *Let $\lambda_k(t)$ be the rate at which new falsehoods are generated in network N_k at time t , influenced by network dynamics and external factors. The number of new falsehoods generated in network N_k at time t , denoted by $X_k(t)$, follows a Poisson distribution:*

$$X_k(t) \sim \text{Poisson}(\lambda_k(t)) \quad (17)$$

Proposition 2.8. Cross-Network Detection. *Let $d_{jk}(t)$ be the probability that a falsehood in network N_k is detected by an actor in network N_j at time t . The number of falsehoods detected and removed from network N_k by network N_j at time t , denoted by $Y_{jk}(t)$, follows a Binomial distribution:*

$$Y_{jk}(t) \sim \text{Binomial}(F_k(t), d_{jk}(t)) \quad (18)$$

Proposition 2.9. Internal Dynamics. *Let $p_k(t)$ be the probability that a true statement in network N_k becomes false at time t , and $q_k(t)$ be the probability that a false statement in network N_k is corrected to become true at time t . The number of true statements becoming false, $Z_k(t)$, and false statements becoming true, $W_k(t)$, in network N_k at time t follow Binomial distributions:*

$$Z_k(t) \sim \text{Binomial}(T_k(t), p_k(t)) \quad (19)$$

$$W_k(t) \sim \text{Binomial}(F_k(t), q_k(t)) \quad (20)$$

The evolution of the network states can be described by the following stochastic difference equations:

$$T_k(t+1) = T_k(t) - Z_k(t) + W_k(t) \quad (21)$$

$$F_k(t+1) = F_k(t) + X_k(t) + Z_k(t) - W_k(t) - Y_{jk}(t) \quad (22)$$

Theorem 2.2. Expected Equilibrium State. *The expected equilibrium state of network N_k is given by:*

$$E[T_k^*] = \frac{q_k^* n_k}{p_k^* + q_k^*} - \frac{\lambda_k^* n_k}{d_{jk}^* (p_k^* + q_k^*)} \quad (23)$$

$$E[F_k^*] = \frac{\lambda_k^* n_k}{d_{jk}^*} \quad (24)$$

where $n_k = E[T_k^*] + E[F_k^*]$ is the expected total number of statements in network N_k at equilibrium, and $\lambda_k^*, d_{jk}^*, p_k^*, q_k^*$ are the long-term average values of the respective time-dependent parameters.

Proof. Step 1: Equilibrium Conditions

At equilibrium, the expected changes in true and false statements should be zero:

$$E[Z_k] = E[W_k] \quad (25)$$

$$E[X_k] = E[Y_{jk}] + E[W_k] - E[Z_k] \quad (26)$$

Step 2: Expected Values and Distribution Justifications

Express the expected values for each random variable, with detailed justifications for the choice of distributions:

$$E[X_k] = \lambda_k^* \quad (\text{Poisson distribution mean}) \quad (27)$$

$$E[Y_{jk}] = F_k^* \cdot d_{jk}^* \quad (\text{Binomial distribution mean}) \quad (28)$$

$$E[Z_k] = T_k^* \cdot p_k^* \quad (\text{Binomial distribution mean}) \quad (29)$$

$$E[W_k] = F_k^* \cdot q_k^* \quad (\text{Binomial distribution mean}) \quad (30)$$

1. $E[X_k] = \lambda_k^*$ (Poisson distribution mean): - The Poisson distribution is used to model the number of events occurring in a fixed interval of time or space, given a known average rate. In this case, X_k represents the number of new falsehoods generated in network N_k . The parameter λ_k^* is the average rate of falsehood generation at equilibrium.

2. $E[Y_{jk}] = F_k^* \cdot d_{jk}^*$ (Binomial distribution mean): - Y_{jk} represents the number of falsehoods in network N_k detected by network N_j . This follows a Binomial distribution because each false statement in N_k has a fixed probability d_{jk}^* of being detected by N_j .

3. $E[Z_k] = T_k^* \cdot p_k^*$ (Binomial distribution mean): - Z_k represents the number of true statements in network N_k that become false. This also follows a Binomial distribution because each true statement has a fixed probability p_k^* of becoming false.

4. $E[W_k] = F_k^* \cdot q_k^*$ (Binomial distribution mean): - W_k represents the number of false statements in network N_k that are corrected to become true. This follows a Binomial distribution as well, with F_k^* false statements each having a probability q_k^* of being corrected.

Step 3: Substitute Expected Values

Substitute these into our equilibrium equations:

$$T_k^* \cdot p_k^* = F_k^* \cdot q_k^* \quad (31)$$

$$\lambda_k^* = F_k^* \cdot d_{jk}^* + F_k^* \cdot q_k^* - T_k^* \cdot p_k^* \quad (32)$$

Step 4: Solve for F_k^* in terms of T_k^*

From Equation 31:

$$F_k^* = T_k^* \cdot \frac{p_k^*}{q_k^*} \quad (33)$$

Step 5: Total Statements

The total number of statements n_k is the sum of true and false statements:

$$n_k = T_k^* + F_k^* \quad (34)$$

$$= T_k^* + T_k^* \cdot \frac{p_k^*}{q_k^*} \quad (35)$$

$$= T_k^* \cdot \left(1 + \frac{p_k^*}{q_k^*}\right) \quad (36)$$

$$= T_k^* \cdot \frac{p_k^* + q_k^*}{q_k^*} \quad (37)$$

Step 6: Solve for T_k^*

Rearranging the last equation from Step 5:

$$T_k^* = n_k \cdot \frac{q_k^*}{p_k^* + q_k^*} \quad (38)$$

Step 7: Solve for F_k^*

Substitute Equation 38 into Equation 33:

$$F_k^* = T_k^* \cdot \frac{p_k^*}{q_k^*} \quad (39)$$

$$= n_k \cdot \frac{q_k^*}{p_k^* + q_k^*} \cdot \frac{p_k^*}{q_k^*} \quad (40)$$

$$= n_k \cdot \frac{p_k^*}{p_k^* + q_k^*} \quad (41)$$

Step 8: Use Equilibrium False Equation

From Equation 32, substitute the expressions for F_k^* and T_k^* :

$$\lambda_k^* = F_k^* \cdot d_{jk}^* + F_k^* \cdot q_k^* - T_k^* \cdot p_k^* \quad (42)$$

$$= n_k \cdot \frac{p_k^*}{p_k^* + q_k^*} \cdot d_{jk}^* + n_k \cdot \frac{p_k^*}{p_k^* + q_k^*} \cdot q_k^* - n_k \cdot \frac{q_k^*}{p_k^* + q_k^*} \cdot p_k^* \quad (43)$$

Here, we clearly show the intermediate steps of substitution.

Step 9: Simplify

Combine terms and simplify, showing each algebraic step:

$$\lambda_k^* = n_k \cdot \frac{p_k^* d_{jk}^* + p_k^* q_k^* - q_k^* p_k^*}{p_k^* + q_k^*} \quad (44)$$

$$= n_k \cdot \frac{p_k^* (d_{jk}^* + q_k^* - q_k^*)}{p_k^* + q_k^*} \quad (45)$$

$$= n_k \cdot \frac{p_k^* d_{jk}^*}{p_k^* + q_k^*} \quad (46)$$

Step 10: Explicit Derivation for F_k^*

Rearrange the last equation to solve for F_k^* :

$$\lambda_k^* = n_k \cdot \frac{p_k^* d_{jk}^*}{p_k^* + q_k^*} \quad (47)$$

$$\Rightarrow F_k^* = \frac{\lambda_k^* \cdot (p_k^* + q_k^*)}{d_{jk}^*} \quad (48)$$

Step 11: Final T_k^*

Subtract F_k^* from n_k to get the final expression for T_k^* :

$$T_k^* = n_k - F_k^* \quad (49)$$

$$= n_k - \frac{\lambda_k^* \cdot (p_k^* + q_k^*)}{d_{jk}^*} \quad (50)$$

$$= n_k \cdot \left(1 - \frac{\lambda_k^*}{d_{jk}^*} \right) \quad (51)$$

$$= \frac{q_k^* \cdot n_k}{p_k^* + q_k^*} - \frac{\lambda_k^* \cdot n_k}{d_{jk}^* \cdot (p_k^* + q_k^*)} \quad (52)$$

This completes the proof, deriving the expressions for $E[T_k^*]$ and $E[F_k^*]$ as stated in the theorem. $\square$

Interpretation and Analysis:

1. Falsehood Generation vs. Detection: The ratio $\frac{\lambda_k^*}{d_{jk}^*}$ represents the balance between falsehood generation and cross-network detection. A higher ratio leads to more false statements at equilibrium.

2. Network Size Effect: Larger networks (higher n_k) can sustain more true and false statements at equilibrium. The number of false statements is now proportional to network size, reflecting increased opportunities for falsehood generation in larger discourse spaces.

3. Internal Dynamics: The ratio $\frac{q_k^*}{p_k^* + q_k^*}$ represents the proportion of non-false statements that are true at equilibrium due to internal dynamics.

4. Cross-Network Efficiency: Higher detection probabilities d_{jk}^* lead to fewer false statements, highlighting the importance of cross-network scrutiny.

5. Asymmetry: Differences in parameters between networks can lead to asymmetric equilibrium states, potentially modeling real-world scenarios where one network is more susceptible to falsehoods or more efficient at detection.

6. Time-Dependent Dynamics: The model now captures how falsehood generation, detection, and internal correction processes may vary over time, allowing for more realistic modeling of network behavior.

7. Stochastic Nature: The use of Poisson and Binomial distributions introduces inherent randomness, reflecting the unpredictable nature of falsehood generation and detection in real-world scenarios.

3 Results

3.1 Numerical Simulations

[TO DO: Read, digest, ensure it is correct.]

This section details the numerical simulations implemented to explore the dynamics of belief systems within a discursive network. The simulations leverage a Markov chain model with dynamically adjusted transition probabilities, reflecting the interaction frequencies and statement validity.

To simulate the interaction counts $N_{s2}(t)$ involving the false statement $s2$ over time, a Poisson distribution is employed. The Poisson distribution is ideally suited for this purpose due to several key characteristics: (i) **Event independence**: The distribution assumes that each interaction is an independent event, a reasonable approximation given the diverse sources and triggers of discussions in a large network, (ii) **fixed average rate**: It is characterized by its rate parameter λ , which simplifies modeling interactions that occur at a consistent average rate over time, (iii) **simplicity and mathematical tractability**: The Poisson distribution provides a straightforward formulation for calculating probabilities of occurrence for various numbers of interactions, facilitating the integration into stochastic models like Markov chains, and (iv) **flexibility in modeling sparse data**: It effectively handles data scenarios where interactions are rare or occur in bursts, typical of communication patterns in social networks.

The Python programming language was used to implement the simulation, employing the NumPy library for handling numerical operations and Matplotlib for visualizing the evolution of belief states over time. Initial states and parameters were set based on theoretical assumptions to reflect realistic interaction dynamics.

The simulation runs over a predefined number of time steps, updating the belief states according to the transition matrix $T(t)$. The stability and convergence of the model are analyzed by observing changes in the proportions of belief states across the simulation period, providing insights into the long-term behavior of the network under varying interaction patterns.

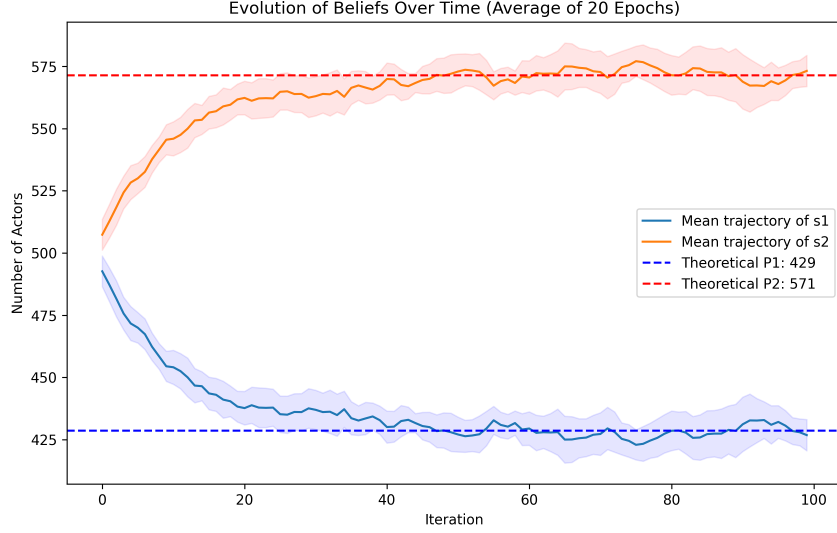


Figure 1: Evolution of beliefs over time, showcasing the dynamic changes in the network based on the specified interaction parameters.

3.2 Invalidation Consensus Algorithm

The following sections detail a methodology for handling requests using multiple Large Language Models (LLMs) to ensure a high reliability and consistency in the responses generated by distributed systems. This approach is critical in scenarios where decision-making requires integration of diverse perspectives or knowledge bases, as often encountered in complex AI systems.

The process initiates with a request which acts as an input requiring processing. This request is then distributed among several instances of LLMs, labeled as LLM_1 to LLM_n . Each of these models processes the request independently and generates respective outputs, denoted as O_1 to O_n . Following the processing, all outputs are collected and assembled for a comparison.

A crucial step in this workflow is the assessment of whether the outputs differ significantly among the models. If significant differences are detected, indicating a lack of consensus, the system injects additional instructions aimed at consensus-building among the models. This might involve recalibration of model parameters or redefinition of input preprocessing. The request is then resent through the system until the outputs converge sufficiently, ensuring consistency. If no significant differences are found during the comparison, the system deems the results consistent enough to formulate a final response based on the collective outputs of the models, thus ending the process.

The algorithm presented in our study is closely aligned with the invalidation model designed to simulate and analyze the dynamics of belief changes within a discursive network, influenced by both true and false statements. This relationship is foundational for understanding the algorithm’s purpose and its operational dynamics within the context of the proposed model. Below, we elaborate on how the algorithm relates to the invalidation

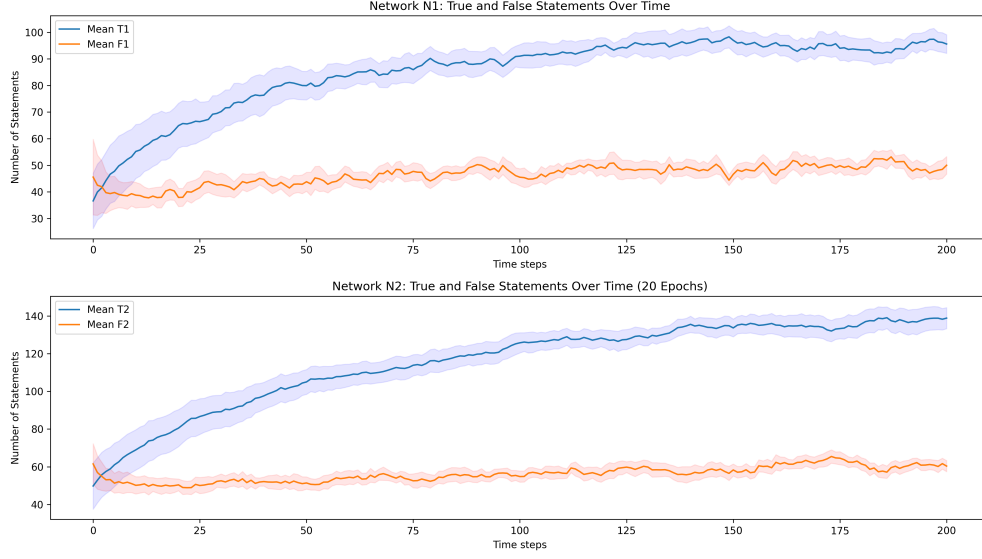


Figure 2: Evolution of beliefs over time, showcasing the dynamic changes in the network based on the specified interaction parameters.

model:

The invalidation model we introduced operates on a Markov chain framework that depicts the stochastic transitions between belief states in a network of actors. These states are significantly influenced by true statements (s_1) and false statements (s_2), wherein the interactions among actors can either reinforce or alter their current beliefs.

3.2.1 Algorithm Steps and Their Functions

1. **Initialization and Distribution:** The process starts by initializing a request, symbolizing the input or stimulus that prompts interaction within the network. This request is distributed among various instances of large language models (LLMs), simulating the diverse responses of actors within the network.
2. **Processing by LLMs:** Each LLM processes the request independently, mirroring the individual actor's processing of information or statements within the network. This independence is crucial for capturing the varied interpretations and reactions among different actors based on their perspectives or biases.
3. **Assembly and Difference Checking:** After processing, the outputs from all LLMs are assembled for comparison. This step is analogous to aggregating different beliefs or reactions within the network to assess consensus or discord.
4. **Consensus and Response Generation:** If significant differences are detected, the algorithm injects consensus instructions and reiterates the process, reflecting the ongoing adjustments and interactions in the network as actors influence each other and gradually move towards a common understanding or belief. If no significant differ-

ences are detected, a response based on collective outputs is generated, signifying the stabilization of beliefs across the network.

3.2.2 Integration with Invalidation Dynamics

The algorithm effectively integrates the dynamics of invalidation by considering how false information (represented by s_2) can influence the network's belief system and how true information (represented by s_1) can be reasserted. The use of LLMs to simulate this process provides a robust framework for exploring the complex interplay of information dissemination, belief formation, and change within large, interactive networks.

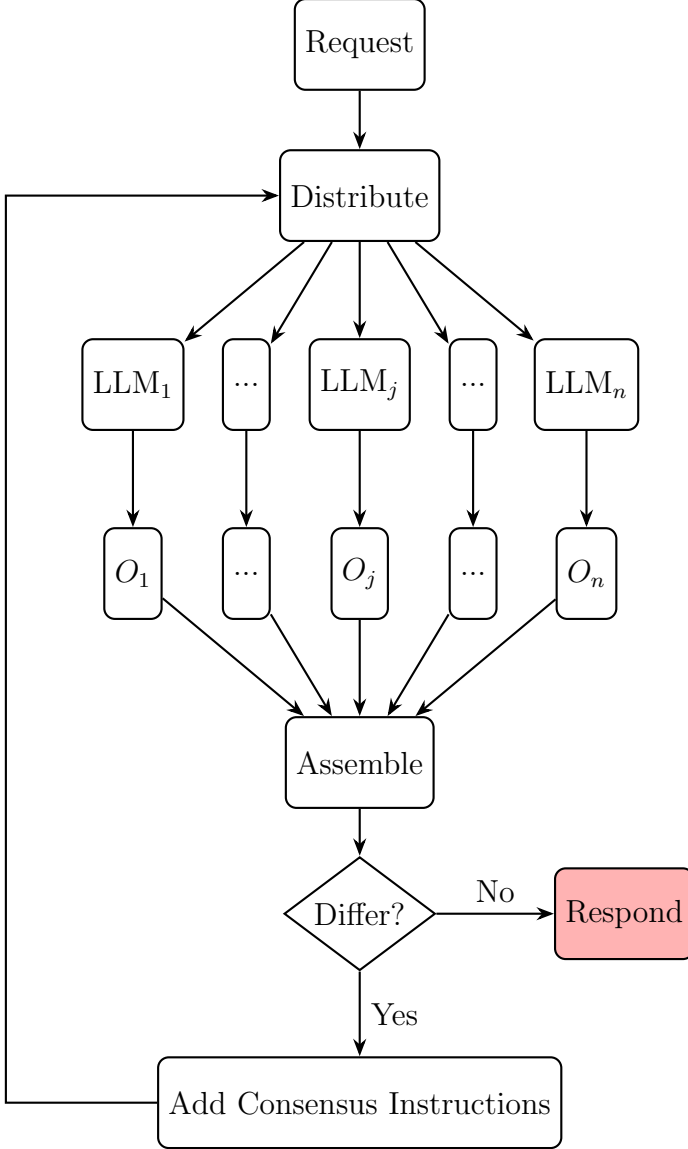
Algorithm 1 Consensus-based Response Generation using Multiple LLMs

Require: Input request

Ensure: Consistent response from multiple LLMs

```

1: Initialize Request
2: Distribute Request to multiple LLMs  $LLM_1, LLM_2, \dots, LLM_n$ 
3: for  $i = 1$  to  $n$  do
4:    $O_i \leftarrow LLM_i(Process(Request))$ 
5: end for
6: Assemble outputs  $O_1, O_2, \dots, O_n$ 
7:  $Difference \leftarrow CheckDifferences(O_1, O_2, \dots, O_n)$ 
8: if  $Difference = \text{True}$  then
9:   Add Consensus Instructions
10:  Go to Step 2
11: else
12:    $Response \leftarrow GenerateResponse(O_1, O_2, \dots, O_n)$ 
13:   Output Response
14: end if
```



4 Conclusion

[TO DO: Clean up conclusions. Ensure that we cite the context in the intro, and address how this approach improves over the state of the art.]

The numerical simulations serve as a powerful tool to understand and predict how beliefs propagate and evolve in a discursive network, illustrating the impact of interaction frequency and statement validity. The use of a Poisson distribution for modeling interaction counts adds a layer of realism and adaptability to the simulations, enhancing their applicability to real-world social dynamics.

To connect the refined model of belief propagation and verification energy to large language models (LLMs), we need to understand how LLMs generate statements and how they can be used to detect falsehoods. LLMs, such as GPT-4, generate text based on patterns they have learned from vast amounts of training data. These models do not have

an inherent understanding of truth or falsehood but instead generate responses that are statistically likely given the input prompt and their training data. Here’s how this relates to our model: LLMs are trained on diverse datasets that include both true (type s_1) and false (type s_2) statements. When generating a response, the LLM samples from this learned distribution, which may include falsehoods if they appear frequently in the training data. In our model, verifying true statements (s_1) requires more energy. Similarly, generating true and accurate information often requires more context, deeper understanding, and reasoning, which the LLM might not always apply if not explicitly prompted to do so. Generating a plausible-sounding but false statement (s_2) might be “easier” in terms of the model’s operations, similar to how s_2 statements deplete less energy. Unlike humans who can verify facts through external sources or reasoning, LLMs rely solely on their training data and prompt context. Without a mechanism to validate information against a ground truth, LLMs may produce statements that are statistically likely but false, akin to our model where falsehoods (s_2) require less verification energy and can propagate more easily.

Using multiple LLMs to spot falsehoods can be more effective than relying on a single model due to several factors that align with our refined model of belief propagation and verification energy. Multiple LLMs trained on different datasets or with different architectures can provide diverse perspectives. When multiple models analyze the same statement, they can cross-verify the information. If one model generates an s_2 statement, other models can provide additional context or challenge the falsehood based on their own learned patterns, similar to increasing the interaction frequency with true statements (s_1) in our model. Relying on a single LLM increases the risk of errors or biases inherent in that model’s training data. By consulting multiple models, we can reduce the likelihood of propagating falsehoods as different models may have different biases and strengths, leading to a more balanced and accurate output. In our model, verification energy depletes quickly when verifying true statements. Multiple LLMs can distribute this verification load. If verifying s_1 depletes energy, having multiple models can effectively pool their verification capacities, ensuring that the energy required to confirm the truth is distributed and managed better. This is analogous to having a collaborative verification process where the burden is shared. When multiple LLMs are used to evaluate a statement, we can derive a consensus based on the majority or weighted agreement. If most models agree that a statement is true or false, this consensus can act as a more reliable indicator than a single model’s output. This is similar to how higher interaction frequencies with true statements (s_1) in our model increase the likelihood of actors adopting s_1 .

By connecting the refined model of belief propagation to the behavior of LLMs, we can better understand the generation and detection of false statements. LLMs generate false statements due to the probabilistic nature of their training data, analogous to how s_2 statements propagate more easily in a network with lower verification energy. Using multiple LLMs for falsehood detection is effective because it distributes the verification load, leverages diverse perspectives, and reduces individual biases, akin to increasing the interaction frequency with true statements and sharing verification energy in our model. This approach aligns with our understanding that collaborative verification and diverse interactions enhance the detection and correction of falsehoods in both human networks and AI systems.

Future research directions could include exploring additional techniques to mitigate invalidation, assessing the effectiveness of existing strategies in real-world applications, and

examining the ethical implications of false information generated by LLMs. Addressing these challenges is essential for the responsible deployment and use of AI technologies.

The parallel between invalidation in LLMs and human discourse opens up new avenues for interdisciplinary research. It challenges us to consider invalidation not as a unique flaw of AI, but as a fundamental aspect of information processing that manifests across different types of cognitive systems. This perspective not only enriches our understanding of AI behavior but also provides a novel lens through which to examine human cognition and social dynamics.

The models presented for invalidaiton propagation provide a comprehensive framework for analyzing the dynamics of falsehood generation, propagation, and detection in isolated and interconnected discursive networks. It allows for the exploration of various scenarios by adjusting parameters such as falsehood generation rates, detection probabilities, and internal correction processes. Future work could focus on extending this model to include more complex network structures, heterogeneous actor behaviors, or the impact of targeted misinformation campaigns. Additionally, empirical studies could help validate the model’s assumptions and refine its parameters for specific real-world contexts.

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